



130 W, 180 fs ultrafast Yb-doped fiber frequency comb based on chirped-pulse fiber amplification

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Abstract: We report on a high-power fiber optical frequency comb consisting of a 250-MHz mode-locked fiber laser and a three-stage cascaded fiber chirped-pulse amplification system. After power scaling, the group velocity dispersion and third-order dispersion, generated in fiber stretcher and amplifiers, are compensated by a grism compressor, outputting a 132-W, 180-fs pulse train. The repetition rate and carrier-envelope offset frequency are locked to a Rb clock with the standard deviations of 1.07 and 0.87 mHz, corresponding to the fractional instability of 8.3×10^{-13} and 1.35×10^{-19} , respectively. Moreover, we investigate the noise characteristics at high average powers, presenting a low-noise property of this high-power fiber OFC.

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1. Introduction

High-power phase-locked near-infrared optical frequency comb has gained significant interest in many applications such as biologic spectra-imaging [1], dual-comb high-resolution spectroscopy [2], and new comb generation in ultraviolet [3] and mid-infrared regions [4]. For frequency comb spectroscopy, high repetition rate must be guaranteed as well as high stability. When targeting ultraviolet and mid-infrared frequency combs, the frequency conversions, such as high harmonic generation [5], optical parametric amplification [6], difference frequency generation [7], demand an origin frequency comb with a high peak power. Therefore, the current power of the available comb sources needs to be further scaled up, realizing a higher peak power together with a high repetition rate. However, ultrahigh power extension of optical frequency comb has always been a huge challenge during the past few years due to nonlinear effects and power instability. As the advances in optical fiber techniques, a large-mode-area double-cladding Yb-doped photonic crystal fiber (LMA DC Yb-PCF) has been widely used to scale up the average power of high-repetition-rate ultrashort pulses [8–10]. Particularly, this kind of fiber possesses a large surface-to-volume ratio, a cladding-pump structure, a broad emission bandwidth, and a large-size core, which can ensure excellent heat dissipation, high pump efficiency, and low nonlinearity, respectively. These advantages synergistically guarantee a generation of high-power phase-stabilized pulses. Moreover, high-power pulses generated by LMA DC Yb-PCF amplifier can maintain the phase stability, which can offer a good carrier-envelope offset (CEO) frequency (f_0) for a low-noise phase locking [11–13].

During pulse amplification, femtosecond pulses will experience nonlinear effects in high-power condition, which will add an extra noise that is hard to be suppressed [14]. An alternative approach to avoiding these nonlinear effects relies on a chirped-pulse amplification (CPA) scheme. In CPA, a huge linear chirp is introduced to optical pulses by a dispersion element before the fiber amplifier. Stretched pulses will go through linear amplification without few effects. Then ultrashort pulses are re-obtained by a compressor owning a reverse dispersion compared with the front stretcher. Generally, the cascaded CPA system indeed promises a power scaling of

high-repetition-rate pulses toward a near-kW level [15]. Typically, a 325-W, 375-fs pulse train has been obtained by a fiber CPA system [16]. However, these high-repetition-rate pulses with average powers up to 100 W have to face a severe problem of pulse distortion mainly coming from the mismatch of both the group velocity dispersion (GVD) and third-order dispersion (TOD) between the stretcher and compressor. To avoid this problem, a fiber CPA system has employed a fiber stretcher, a prism compressor and a grating compressor to match GVD and TOD, emitting a 150-W, 418-fs Gaussian pulse train [17]. Another research team has used the same transmission gratings for both the pulse stretcher and compressor, outputting a pulse train with an average power of 100 W and a pulse duration of 270 fs [18]. In 1997, S. Kane and J. Squier have successfully simulated the management capacity of both GVD and TOD on grism system [19]. Recently, a proposed grism compressor setup has attracted much attention due to its excellent compensation capacity of GVD and TOD [20] in high-power fiber amplification. All these schemes can provide clean ultrashort pulses for the low-noise phase detection and control, especially the easiest grism-based scheme.

In this paper, we demonstrate the generation of high-power femtosecond fiber frequency comb based on a three-stage cascaded fiber CPA system. To obtain low-noise pulses, a pair of grisms is employed to compensate the GVD and TOD generated in fiber stretcher and amplifiers. After the grism compressor, a clean pulse with ~99% power maintained in the main peak is achieved with an average power of 132 W and a pulse duration of 180 fs. Through a self-reference f -2 f interferometer and detection units, we acquire a 250 MHz repetition rate frequency signal (f_r) and a 20 MHz beating signal (f_0). Finally, the f_r and f_0 are locked to a stabilized Rb clock with a frequency standard deviation of 1.07 and 0.87 mHz, respectively. For a 1000 s averaging time, the corresponding fractional instability of f_r and f_0 are 8.3×10^{-13} and 1.35×10^{-19} . Moreover, we investigate the noise characteristics of both f_r and f_0 , further verifying the stability of the high-power frequency comb. To best of our knowledge, this is the highest average power ever reported for a CPA fiber comb system, which can provide an excellent high-power source for the generation of high-power ultraviolet and mid-infrared OFC.

2. Experimental setup

The configuration for the high-power fiber frequency comb system is shown in Fig. 1, including a nonlinear-polarization-rotation (NPR) mode-locked seed laser, a fiber stretcher, three fiber amplifiers, and a grism compressor. The homemade ring oscillator consisted of a 15-cm Yb-doped fiber (YDF, YB401, Coractive), a 45-cm single-mode fiber (SMF, HI1060) and a 30-cm spatial structure, corresponding to a repetition rate of 250 MHz. A pair of transmission gratings (groove density: 1000 line/mm) was employed to manage the net cavity dispersion at a near-zero dispersion where the oscillator operated at the stretched-pulse regime and had the lowest noise [21]. Both SMF and YDF in cavity were mounted on a copper plate held at 23 °C (Room temperature of laboratory) by a thermoelectric cooler (TEC) module. To isolate environmental disturbance, the whole cavity was surrounded by a heat insulation material together with sound insulation cotton and two high-density aluminum plates. For the locking of the repetition rate, a piezo actuator (PZT) was mounted on an intracavity zero-degree reflection mirror. After propagating in a 180-m SMF stretcher, the 40-mW 50-fs seed pulses were sufficiently stretched for a linear CPA amplification in next three cascaded fiber amplifiers.

A single-mode YDF with a length of 20 cm was chosen as a gain media for the first amplifier. When pumped by a 650-mW, 976-nm laser, an average power of 280 mW was obtained. Then, the second amplifier, pumped by a 20-W semiconductor laser at 976 nm, scaled the average power up to ~12 W with an optical-to-optical efficiency of 60%. After the third power amplifier, the pulses were amplified to 268 W with a total of 365-W pump light delivered into the gain fiber. Specifically, the second and third amplifiers were comprised of two polarization maintain (PM) LMA Yb-PCFs with a length of 1.5 m and 2 m, respectively, which had a maximal pump

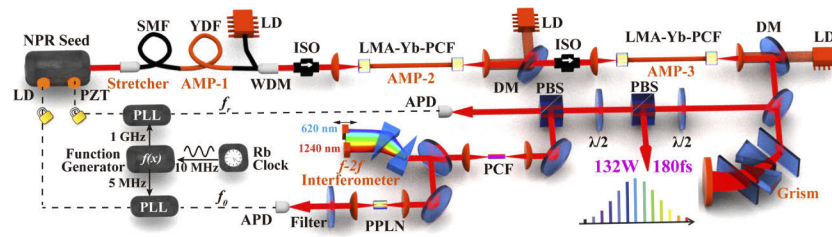


Fig. 1. Experimental setup. NPR: nonlinear polarization rotation; PZT: piezo actuator; SMF: single-mode fiber; YDF: Yb-doped fiber; LD: laser diode; WDM: wavelength-division multiplexing; ISO: optical isolator; LMA Yb-PCF: large-mode-area Yb-doped photonic crystal fiber; DM: dichroic mirror; $\lambda/2$: half wave plate; PBS: polarization beam splitter; PPLN: periodically poled lithium niobate crystal; APD: avalanche photodiode; PLL: phase locking loop; f_r , f_0 : repetition rate frequency, carrier-envelope offset frequency.

absorption of 10 dB/m at 976 nm. These PM LMA Yb-PCFs possessed a double-cladding structure with a 40- μm core and a 200- μm inner cladding, and were mounted on a copper plate held at 18 °C by a water-cooling setup. Moreover, all the end faces of the PM LMA Yb-PCFs were polished at a slant angle of 8 ° to minimize an end reflection. For a high pump efficiency, all these fiber amplifiers used a backward pumping construction. And two additional optical isolators were inserted to avoid damage from the backward reflective light and the unabsorbed pumping light. Next, amplified pulses were compressed by a double-pass grism compressor that consisted of a pair of transmission gratings (groove density: 1000 line/mm) and a pair of equilateral prisms (SF10). Experimentally, the grating pair and prism pair were placed in parallel with a Littrow angle and Brewster angle, respectively, which facilitated a high efficiency and a good-quality beam. Just by adjusting the separation of the gratings and insertion of prisms, we changed the secondary and tertiary dispersion to compensate the GVD and TOD of amplified pulses.

Then, the compressed pulses were divided into three parts by two polarization beam splitter (PBS). A 40-mW part was launched at a photoelectric detector for a detection of f_r . Another part of 500 mW was delivered into a 15-cm high nonlinear photonic crystal fiber (HN-PCF, 3.3-890) for a generation of super-continuum octave spectrum. Two peaks at 620 nm and 1240 nm in super-continuum octave spectrum were filtered and imported to a self-reference $f-2f$ interferometer for a detection of f_0 . After that, f_r and f_0 signals were inputted to two phase locked loops, which drove the PZT and pump current in cavity to lock the repetition rate and CEO frequency, respectively. Finally, a phase-stabilized frequency comb with an average power of 132 W was obtained.

3. Results and discussions

To characterize this high-power CPA fiber comb, we measured its temporal, spectral, and power properties via a commercial autocorrelator (Pulse Check, APE), an optical spectrum analyzer (AQ6370, Yokogawa), and a power meter (PM1K, Coherent), respectively. As shown in Fig. 2(a), the output power of the CPA system increases with a steady optical-to-optical efficiency around ~73.5% when increasing pump power. A maximal average power of 268 W is achieved while a 365-W pump light is delivered into the third amplifier, exhibiting an outstanding scaling capability of high-repetition-rate pulses. Then amplified pulses are compressed by a grism-based compressor with a maximal output power of 132 W, as described in Fig. 2(b). Moreover, the grating and prism were placed with an incidence angle of 29.8 ° (Littrow angle) and 59.8 ° (Brewster angle), respectively, which facilitated a double-pass efficiency of ~50%. Figures 2(c) and 2(d) show the spectral and temporal characteristics of pulses. Due to the effective

compensations for both GVD and TOD in the compressor, the pulses are compressed to 180 fs with $\sim 99\%$ power contained in the main pulse, and a corresponding wavelength is centered at 1038 nm with a half bandwidth of ~ 10 nm. Compared with the seed spectrum with a 45-nm bandwidth, an obvious spectral narrowing, mainly caused by gain narrowing effect, is observed in Fig. 2(c). Moreover, the clean pulse profile shown in Fig. 2(d) reveals that the grism compressor has an excellent dispersion management capability. These clean pulses are conducive to obtain the low-noise f_0 , contributing to a long-term stability frequency comb.

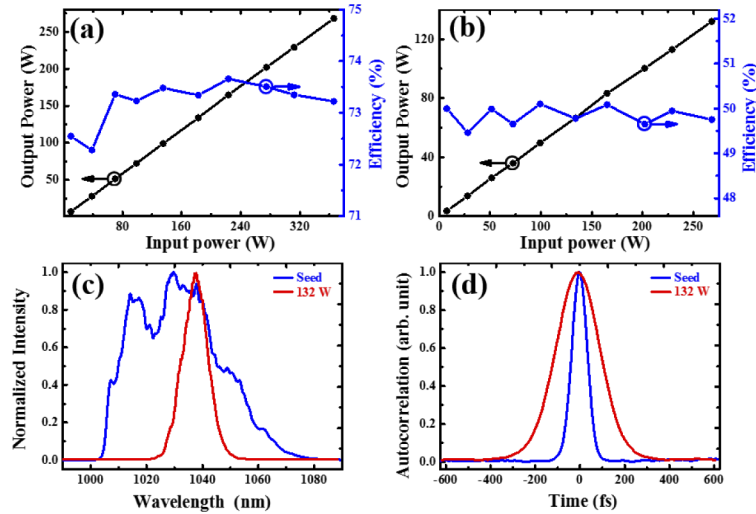


Fig. 2. Output power (black line) and efficiency (blue line) of CPA system versus launched pump power. (b) Output power (black line) and efficiency (blue line) of grism compressor depending on input power. (c) Spectral properties of NPR seed pulses (black line) and 132-W CPA pulses (red line). (d) Temporal properties of NPR seed pulses (black line) and 132-W CPA pulses (red line).

After linear amplification, a 15-cm-long HN-PCF together with a classical self-reference $f-2f$ interferometer, and an avalanche photodiode (APD) were employed to produce a beating signal of f_0 while another APD was used to detect the f_r signal. As depicted in Fig. 3(a), a signal analyzer (N9010A, Agilent) was chosen to measure the radio frequency spectrum of beating signal and repetition rate, achieving a 20-MHz f_0 and a 250-MHz f_r , with the signal-to-noise ratios (SNR) of 30 and 65 dB, respectively. At the max power of 132 W, we counted both the locking f_r and $f_0/2$ signals using a digital counter (5313A, Agilent) with a 1 s gate time. We separately describe the frequency differences of f_r and f_0 as a function of measurement time in Figs. 3(b) and 3(c). For a total of 1000 s counting time, the f_r varies among ± 5 mHz (Center frequency: 250 MHz) with a corresponding standard deviation of 1.07 mHz. Similar to f_r , the $f_0/2$ is locked to ± 2 mHz (Center frequency: 10 MHz) with the same-level standard deviation of 0.87 mHz. Figures 4(a) and 4(b) show the Allan deviations calculated from the time traces of f_r and $f_0/2$, respectively. For a 1000 s averaging time, a fractional statistical uncertainty of f_r is 8.3×10^{-13} while that of f_0 is 1.35×10^{-19} . The slopes are proportional to $\tau^{-1/2}$, confirming the characteristics of the Allan deviations with white frequency noise [22–24]. These fractional statistical uncertainties are mainly limited by the uncertainty of reference signal synchronized by a commercial Rb clock (FS725, Standard Research Systems). The locked of f_r and f_0 provide a direct proof for the generation of a high-power frequency comb. These deviations and uncertainties also certify that the comb inherited a frequency stability from Rb clock.

To investigate the noise properties of the high-power optical frequency comb, a commercial phase noise analyzer (ROHDE & SCHWARZ) was used to measure the intensity and phase

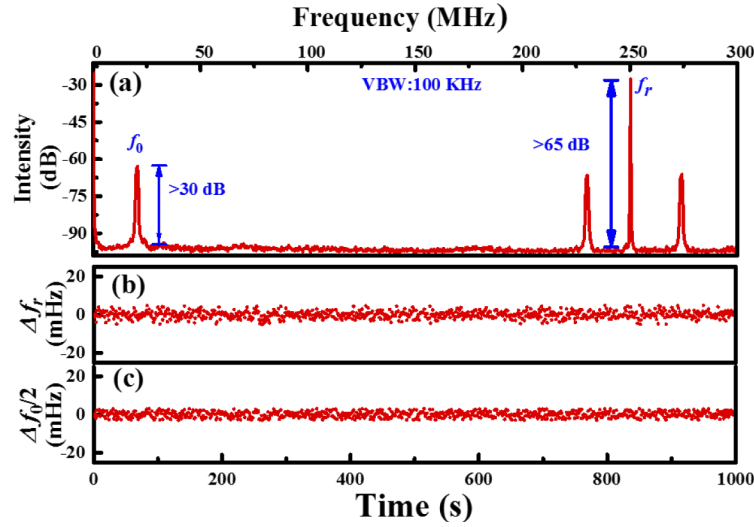


Fig. 3. (a) Radio frequency spectrum of a self-reference f - $2f$ interferometer with f_r and f_0 at 250 MHz and 20 MHz, respectively. Difference of f_r (b) and f_0 (c) as a function of measurement time.

noises of locked f_r and f_0 . Specific noise characteristics of f_0 at the pulse powers of 40, 80, and 132 W are shown in Fig. 5. As seen in Fig. 5(a), with the power increasing, phase noise degenerates a little, but still maintains a flat noise trace in the low frequency region. Therefore, it is obvious that this CPA fiber system can guarantee a low-phase-noise power scaling for f_0 . However, some issues such as temperature variation, gain narrowing, and pump wavelength shifting can't be completely removed, directly resulting in a weak power-related disturbance. When pulses are delivered into HN-PCF for a super-continuum generation and f_0 detection, the disturbance will introduce intensity noise. Thus, the intensity noise of f_0 sharply increases with the power increasing, but is still in a relatively low-noise condition considering such high power, as described in Fig. 5(b). To further characterize the noises, the traces are integrated from 3 MHz to 1 Hz, as exhibited in Figs. 5(c) and 5(d). Integrated phase noises at different pulse powers of 40, 80, and 132 W are 0.172, 0.207, and 0.265 rad, corresponding to integrated intensity noises of 0.038%, 0.23% and 1.7%, respectively. All these noises and parameters reveal that the f_0 is completely controlled, and is effectively locked at a relatively low noise condition.

On the other hand, the repetition rate signal was directly detected by a photoelectric detector with 40-mW pulses imported, then a 250-MHz f_r was obtained after filtered by a circuit filter. On

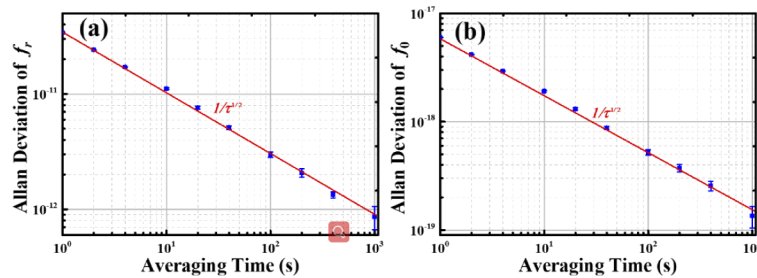


Fig. 4. Allan deviation plots calculated from the time traces of f_r (a) and f_0 (b). A counter gate time (τ) of 1 s was used in the measurements.

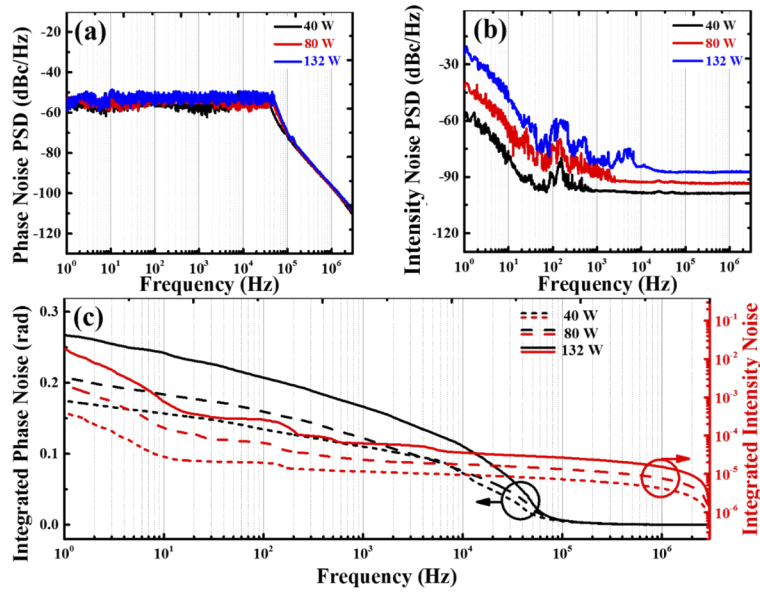


Fig. 5. Phase (a) and intensity (b) noises of locked f_0 measured at a power of 40 (black line), 80 (red line), and 132 W (blue line). (c) Integrated phase (black) and intensity (red) noises of locked f_0 measured at the powers of 40 (short dash), 80 (dash), and 132 W (line).

account of the direct detection without the $f-2f$ interferometer, the noise of f_r is more insensitive to the power-related disturbances than the that of f_0 . As we can see in Fig. 6(a), the phase noise hardly increases with the power rising, revealing that this amplification has little impacts

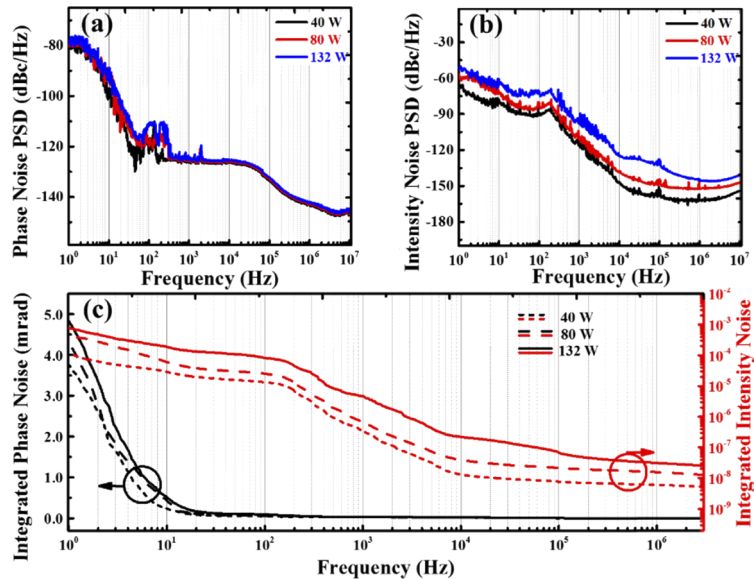


Fig. 6. Phase (a) and intensity (b) noises of locked f_r at a power of 40 (black line), 80 (red line), and 132 W (blue line). (c) Integrated phase (black) and intensity (red) noises of locked f_r at the powers of 40 W (short dash), 80 (dash), and 132 W (line).

on the phase noise of f_r . However, a power instability increases with the power scaling, which leads to direct disturbances on the intensity noise of f_r . Typical degradation of intensity noise is clearly observed with the power increasing, as shown in Fig. 6(b). Especially, several noise peaks around 300 Hz are found in Figs. 5(b), 6(a) and 6(b), which are introduced by a PZT circuit with the same driving frequency of 300 Hz. After an integration from 10 MHz to 1 Hz, we present an integrated phase and intensity noises at different output powers, as depicted in Fig. 6(c). Integrated phase noises of f_r at different pulse powers of 40, 80, and 132 W are 3.7, 4.2, and 4.8 mrad, corresponding to integrated intensity noises of 0.011%, 0.041%, and 0.076%, respectively. These noise parameters demonstrate an effective locking of f_r , verifying that the repetition rate of this high-power CPA-based frequency comb is controlled at a low noise level.

4. Summary

In summary, we report on the generation of high-power femtosecond fiber frequency comb based on a three-stage cascaded CPA system and a NPR mode-locked Yb-doped fiber laser. Power scaling of pulses is carried out through one SMF amplifier and two LMA DC Yb-PCF amplifiers, with a cascade structure. After the compensation for GVD and TOD in grism compressor, amplified pulses are compressed to 180 fs with an average power of >130 W. Through self-reference f - $2f$ interferometer and detection units, a 250-MHz repetition rate frequency signal (f_r) and a 20-MHz beating signal (f_0) are acquired. And both the f_r and f_0 are locked to Rb clock with a frequency standard deviation of 1.07 and 0.87 mHz, which corresponds to fractional statistical uncertainties of 8.3×10^{-13} and 1.35×10^{-19} for a 1000 s averaging time, respectively. Moreover, we investigate the noise characteristics of locked f_r and f_0 , further proving that this comb is operating at low-noise level. This kind of high-power comb can be employed as an optical source for the generation of the ultraviolet and mid-infrared comb, precision spectroscopy, and frequency metrology.

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Disclosures

The authors declare no conflicts of interest.

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